



Jayesh Khalane

B. Tech, Information Technology 2022-26
JSPM's Rajarshi Shahu College of Engineering, Pune
Affiliated to Savitribai Phule Pune University

PROFILE

Driven by a passion for problem-solving, I actively seek opportunities to expand my skill set and leverage new tools to overcome challenges. My commitment to continuous learning has enabled me to contribute to diverse projects, where I focused on improving operational efficiency and creating practical, innovative solutions

CONTACT

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WEBSITE

[GitHub](#)
[LinkedIn](#)

LANGUAGE

English
Hindi
Marathi
German

HOBBIES

Reading
Chess
Swimming

WORK EXPERIENCE

• **E-yantra -IIT Bombay** [May 27th 2024 - July 19th 2024]
Worked on TurtleBot-based swarm project at ARMS Lab, IIT Bombay, under Prof. Leena Vachhani and PhD mentor Sweksha Jain. Designed and implemented a novel communication-less multi-agent identification model enabling decentralized clustering without localization only using 2D LiDAR data as input. Published as “**Guided Swarm Self-Clustering in Safe Area**” at the Conference: 2025 IEEE/SICE International Symposium on System Integration (SII), Munich Germany.

• **Technische Universität Berlin, Germany** [June 2025 - Aug 2025]
Engineering Trainee
Led and contributed to projects on *Industry 4.0 Integrating Sustainability*, applying IoT integration, automation, and lifecycle analysis. Engaged in cross-cultural teamwork, industry-focused workshops, and problem-solving sessions addressing real-world sustainability challenges.

• **CiPHER ROBOCON Team,** [Sep 2023 - Aug 2024]
Worked on autonomous robots involving LiDAR mapping, motion planning, localization, and navigation. Built pick-and-place mechanisms and integrated on board computers, and LiDAR sensors for real-time perception and control.

PROJECTS

• **Astro Tinker Robot e-YRC IIT-B | Team Lead** [Sep 5th 2023 - Feb 27th 2024]
Worked on FPGA board to implement custom RISC-V 32-bit I-type processor, serial communication, PWM generation, pulse detection, UART, robust line following, pick up and place mechanism. Designed a complete robotic solution to the given problem statement by IIT-B robotic competition.

• **Capstone Project | Team Lead** [Dec 15th 2024 - ongoing]
Learning Based Methods for large scale 3D reconstruction from Satellite Imagery. Explored deep learning and computer vision techniques for generating high-fidelity 3D models from satellite data like NeRF, MVS-NET, G-Splats and SV-Raster. Improved Existing COLMAP pipeline to reduce noise in reconstruction.

• **Developed RAID for Network Attached Storage.** [FY Project]
• **Distributed computing raspberry-Pi cluster** [SY Project]

EDUCATION

Pratibha International School Pune	SSC	2008-2020	77.20%
VVNP Jr college Pune	HSC	2020-2022	78.17%
JSPM'S RSCOE Pune	B Tech	2022-2026	8.78 CGPA

SKILLS

Programming: C, C++, Python, Bash, Verilog
Interest: OS, Hardware, Computer Networks, Embedded systems.